

Symplectic topology and Toric Varieties

Part II: Hamiltonian geometry and Delzant's Theorem

Abdullah Ahmed¹

¹Department of Mathematics
Lahore University of Management Sciences, Pakistan
Supervisor: Dr. Shaheen Nazir

UMD JAMS Student Seminar, November 21, 2024

Table of Contents

- 1 Review of symplectic structure
- 2 Hamiltonian actions
- 3 Generalisation through Lie Theory
- 4 Delzant's Theorem

Table of Contents

- 1 Review of symplectic structure
- 2 Hamiltonian actions
- 3 Generalisation through Lie Theory
- 4 Delzant's Theorem

Important definitions

- A symplectic manifold (M, ω) is an even-dimensional smooth manifold equipped with a non-degenerate, closed 2-form ω .

Important definitions

- A symplectic manifold (M, ω) is an even-dimensional smooth manifold equipped with a non-degenerate, closed 2-form ω .
- For M compact, given any vector field $X : M \rightarrow TM$, there exists a unique family of diffeomorphisms denoted by $\exp(tX)$ called the *flow* such that $\exp(tX)|_{t=0} = \text{id}_M$ and $\frac{d}{dt}\exp(tX)(p) = X_{\exp(tX)(p)}$.

Important definitions

- A symplectic manifold (M, ω) is an even-dimensional smooth manifold equipped with a non-degenerate, closed 2-form ω .
- For M compact, given any vector field $X : M \rightarrow TM$, there exists a unique family of diffeomorphisms denoted by $\exp(tX)$ called the *flow* such that $\exp(tX)|_{t=0} = \text{id}_M$ and $\frac{d}{dt}\exp(tX)(p) = X_{\exp(tX)(p)}$.
- The flows passing through a specific initial point $p \in M$ are called *integral curves*, and form a group under composition.

Important definitions

- A symplectic manifold (M, ω) is an even-dimensional smooth manifold equipped with a non-degenerate, closed 2-form ω .
- For M compact, given any vector field $X : M \rightarrow TM$, there exists a unique family of diffeomorphisms denoted by $\exp(tX)$ called the *flow* such that $\exp(tX)|_{t=0} = \text{id}_M$ and $\frac{d}{dt}\exp(tX)(p) = X_{\exp(tX)(p)}$.
- The flows passing through a specific initial point $p \in M$ are called *integral curves*, and form a group under composition.
- The Lie derivative is an operator defined on differential forms which measures change along a vector field: $\mathfrak{L}_X \omega = \frac{d}{dt}\exp(tX)^* \omega|_{t=0}$.

Important definitions

- A symplectic manifold (M, ω) is an even-dimensional smooth manifold equipped with a non-degenerate, closed 2-form ω .
- For M compact, given any vector field $X : M \rightarrow TM$, there exists a unique family of diffeomorphisms denoted by $\exp(tX)$ called the *flow* such that $\exp(tX)|_{t=0} = \text{id}_M$ and $\frac{d}{dt}\exp(tX)(p) = X_{\exp(tX)(p)}$.
- The flows passing through a specific initial point $p \in M$ are called *integral curves*, and form a group under composition.
- The Lie derivative is an operator defined on differential forms which measures change along a vector field: $\mathcal{L}_X\omega = \frac{d}{dt}\exp(tX)^*\omega|_{t=0}$.
- (Cartan's formula) $\mathcal{L}_X\omega = di_X\omega + i_Xd\omega$, where $i_X\omega = \omega(X, \cdot)$ is a 1-form.

Important definitions

- A symplectic manifold (M, ω) is an even-dimensional smooth manifold equipped with a non-degenerate, closed 2-form ω .
- For M compact, given any vector field $X : M \rightarrow TM$, there exists a unique family of diffeomorphisms denoted by $\exp(tX)$ called the *flow* such that $\exp(tX)|_{t=0} = \text{id}_M$ and $\frac{d}{dt}\exp(tX)(p) = X_{\exp(tX)(p)}$.
- The flows passing through a specific initial point $p \in M$ are called *integral curves*, and form a group under composition.
- The Lie derivative is an operator defined on differential forms which measures change along a vector field: $\mathcal{L}_X\omega = \frac{d}{dt}\exp(tX)^*\omega|_{t=0}$.
- (Cartan's formula) $\mathcal{L}_X\omega = di_X\omega + i_Xd\omega$, where $i_X\omega = \omega(X, \cdot)$ is a 1-form.
- The Lie bracket $W = [X, Y]$ of two vector fields X, Y is the unique vector field W such that $W_p f = X_p Y f - Y_p X f$ for all p and $f \in C^\infty(M)$.

Table of Contents

1 Review of symplectic structure

2 Hamiltonian actions

3 Generalisation through Lie Theory

4 Delzant's Theorem

Symplectic and Hamiltonian vector fields

Definition

A vector field X on a symplectic manifold (M, ω) is called symplectic if it preserves ω in the sense that $\mathcal{L}_X \omega = 0$.

Symplectic and Hamiltonian vector fields

Definition

A vector field X on a symplectic manifold (M, ω) is called symplectic if it preserves ω in the sense that $\mathfrak{L}_X \omega = 0$.

- Using the Cartan homotopy formula: $\mathfrak{L}_X \omega = di_X \omega + i_X d\omega$, an equivalent criterion is $di_X \omega = 0$.

Symplectic and Hamiltonian vector fields

Definition

A vector field X on a symplectic manifold (M, ω) is called symplectic if it preserves ω in the sense that $\mathfrak{L}_X \omega = 0$.

- Using the Cartan homotopy formula: $\mathfrak{L}_X \omega = di_X \omega + i_X d\omega$, an equivalent criterion is $di_X \omega = 0$.
- Example: $X = \frac{\partial}{\partial \theta_i}$ is symplectic on $(\mathbb{T}^2, d\theta_1 \wedge d\theta_2)$.

Symplectic and Hamiltonian vector fields

Definition

A vector field X on a symplectic manifold (M, ω) is called symplectic if it preserves ω in the sense that $\mathfrak{L}_X \omega = 0$.

- Using the Cartan homotopy formula: $\mathfrak{L}_X \omega = di_X \omega + i_X d\omega$, an equivalent criterion is $di_X \omega = 0$.
- Example: $X = \frac{\partial}{\partial \theta_i}$ is symplectic on $(\mathbb{T}^2, d\theta_1 \wedge d\theta_2)$.
- Imposing an extra condition, we produce Hamiltonian vector fields:

Symplectic and Hamiltonian vector fields

Definition

A vector field X on a symplectic manifold (M, ω) is called symplectic if it preserves ω in the sense that $\mathcal{L}_X \omega = 0$.

- Using the Cartan homotopy formula: $\mathcal{L}_X \omega = di_X \omega + i_X d\omega$, an equivalent criterion is $di_X \omega = 0$.
- Example: $X = \frac{\partial}{\partial \theta_i}$ is symplectic on $(\mathbb{T}^2, d\theta_1 \wedge d\theta_2)$.
- Imposing an extra condition, we produce Hamiltonian vector fields:

Definition

A vector field X_H on a symplectic manifold (M, ω) is called Hamiltonian if $i_X \omega = dH$ is exact. H is then called the Hamiltonian function of X .

Symplectic and Hamiltonian vector fields

- By non-degeneracy of ω , every $H \in C^\infty(M)$ has a corresponding Hamiltonian vector field.

Symplectic and Hamiltonian vector fields

- By non-degeneracy of ω , every $H \in C^\infty(M)$ has a corresponding Hamiltonian vector field.
- Hamiltonian functions are preserved by their respective Hamiltonian vector fields. (Cartan formula)

Symplectic and Hamiltonian vector fields

- By non-degeneracy of ω , every $H \in C^\infty(M)$ has a corresponding Hamiltonian vector field.
- Hamiltonian functions are preserved by their respective Hamiltonian vector fields. (Cartan formula)
- Example: Every symplectic vector field is locally Hamiltonian.

Symplectic and Hamiltonian vector fields

- By non-degeneracy of ω , every $H \in C^\infty(M)$ has a corresponding Hamiltonian vector field.
- Hamiltonian functions are preserved by their respective Hamiltonian vector fields. (Cartan formula)
- Example: Every symplectic vector field is locally Hamiltonian.
- Example: $X_h = \frac{\partial}{\partial \theta}$ is Hamiltonian on $(S^2, d\theta \wedge dh)$, where h is the height function.

Symplectic and Hamiltonian vector fields

- By non-degeneracy of ω , every $H \in C^\infty(M)$ has a corresponding Hamiltonian vector field.
- Hamiltonian functions are preserved by their respective Hamiltonian vector fields. (Cartan formula)
- Example: Every symplectic vector field is locally Hamiltonian.
- Example: $X_h = \frac{\partial}{\partial \theta}$ is Hamiltonian on $(S^2, d\theta \wedge dh)$, where h is the height function.
- Example: The exactness condition above reduce to Hamilton's equations on the trajectory of a particle moving in standard phase space $(\mathbb{R}^{2n}, \sum_i dq_i \wedge dp_i)$, with $H : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ as the energy function.

Hamiltonian actions

- The ideas above can be reformulated in terms of $\mathbb{R}$ -actions or S^1 -actions on (M, ω) : $t \mapsto \rho_t \in \text{Diff}(M)$.

Hamiltonian actions

- The ideas above can be reformulated in terms of $\mathbb{R}$ -actions or S^1 -actions on (M, ω) : $t \mapsto \rho_t \subset \text{Diff}(M)$.
- ρ_t can be differentiated to obtain X , of which it is an integral curve.

Hamiltonian actions

- The ideas above can be reformulated in terms of $\mathbb{R}$ -actions or S^1 -actions on (M, ω) : $t \mapsto \rho_t \in \text{Diff}(M)$.
- ρ_t can be differentiated to obtain X , of which it is an integral curve.

Definition

If an $\mathbb{R}$ or S^1 -action has an associated vector field that is either symplectic or Hamiltonian, the action is called symplectic or Hamiltonian respectively.

Hamiltonian actions

- The ideas above can be reformulated in terms of $\mathbb{R}$ -actions or S^1 -actions on (M, ω) : $t \mapsto \rho_t \subset \text{Diff}(M)$.
- ρ_t can be differentiated to obtain X , of which it is an integral curve.

Definition

If an $\mathbb{R}$ or S^1 -action has an associated vector field that is either symplectic or Hamiltonian, the action is called symplectic or Hamiltonian respectively.

- Example: Let $\mathbb{R}$ act on $(\mathbb{R}^{2n}, \omega = \sum_i dx_i \wedge dy_i)$ as $\rho_t(p) = (x_1, \dots, x_n, y_1 - t, y_2, \dots, y_n)$. The corresponding vector field is the constant vector field $X = -\frac{\partial}{\partial y_1}$, with Hamiltonian function x_1 .

Hamiltonian actions

- The ideas above can be reformulated in terms of $\mathbb{R}$ -actions or S^1 -actions on (M, ω) : $t \mapsto \rho_t \subset \text{Diff}(M)$.
- ρ_t can be differentiated to obtain X , of which it is an integral curve.

Definition

If an $\mathbb{R}$ or S^1 -action has an associated vector field that is either symplectic or Hamiltonian, the action is called symplectic or Hamiltonian respectively.

- Example: Let $\mathbb{R}$ act on $(\mathbb{R}^{2n}, \omega = \sum_i dx_i \wedge dy_i)$ as $\rho_t(p) = (x_1, \dots, x_n, y_1 - t, y_2, \dots, y_n)$. The corresponding vector field is the constant vector field $X = -\frac{\partial}{\partial y_1}$, with Hamiltonian function x_1 .
- Let S^1 act on S^2 as $\rho_\theta(\alpha, h) = (\alpha + \theta, h)$. The corresponding vector field is the Hamiltonian vector field $X_h = \frac{\partial}{\partial \theta}$.

Table of Contents

- 1 Review of symplectic structure
- 2 Hamiltonian actions
- 3 Generalisation through Lie Theory**
- 4 Delzant's Theorem

Lie groups and Lie algebras

Definition

A Lie group is a manifold G equipped with a group structure where the multiplication and inverse operations are also smooth.

Lie groups and Lie algebras

Definition

A Lie group is a manifold G equipped with a group structure where the multiplication and inverse operations are also smooth.

- $\mathbb{R}^n$ and $\mathbb{T}^n$ are Lie groups under standard addition.

Lie groups and Lie algebras

Definition

A Lie group is a manifold G equipped with a group structure where the multiplication and inverse operations are also smooth.

- $\mathbb{R}^n$ and $\mathbb{T}^n$ are Lie groups under standard addition.
- Most matrix groups $GL(V)$, $U(n)$, $SU(n)$, $O(n)$ etc are Lie groups.

Lie groups and Lie algebras

Definition

A Lie group is a manifold G equipped with a group structure where the multiplication and inverse operations are also smooth.

- $\mathbb{R}^n$ and $\mathbb{T}^n$ are Lie groups under standard addition.
- Most matrix groups $GL(V)$, $U(n)$, $SU(n)$, $O(n)$ etc are Lie groups.

Definition

A Lie algebra is a vector space $\mathfrak{g}$ equipped with a bilinear, anti-symmetric binary operation $[\cdot, \cdot]$ called the Lie bracket, that satisfies the Jacobi identity $[x, [y, z]] + [z, [x, y]] + [y, [z, x]] = 0$.

Lie groups and Lie algebras

Definition

A Lie group is a manifold G equipped with a group structure where the multiplication and inverse operations are also smooth.

- $\mathbb{R}^n$ and $\mathbb{T}^n$ are Lie groups under standard addition.
- Most matrix groups $GL(V)$, $U(n)$, $SU(n)$, $O(n)$ etc are Lie groups.

Definition

A Lie algebra is a vector space $\mathfrak{g}$ equipped with a bilinear, anti-symmetric binary operation $[\cdot, \cdot]$ called the Lie bracket, that satisfies the Jacobi identity $[x, [y, z]] + [z, [x, y]] + [y, [z, x]] = 0$.

- The vector space $\chi(M)$ of all vector fields on M is a Lie algebra with the Lie bracket $[X, Y]_p f = X_p Yf - Y_p Xf$.

Lie groups and Lie algebras

Definition

A Lie group is a manifold G equipped with a group structure where the multiplication and inverse operations are also smooth.

- $\mathbb{R}^n$ and $\mathbb{T}^n$ are Lie groups under standard addition.
- Most matrix groups $GL(V)$, $U(n)$, $SU(n)$, $O(n)$ etc are Lie groups.

Definition

A Lie algebra is a vector space $\mathfrak{g}$ equipped with a bilinear, anti-symmetric binary operation $[\cdot, \cdot]$ called the Lie bracket, that satisfies the Jacobi identity $[x, [y, z]] + [z, [x, y]] + [y, [z, x]] = 0$.

- The vector space $\chi(M)$ of all vector fields on M is a Lie algebra with the Lie bracket $[X, Y]_p f = X_p Yf - Y_p Xf$.
- $\chi^{ham}(M)$ and $\chi^{symp}(M)$ form Lie-subalgebras of $\chi(M)$.

Lie groups and Lie algebras

Definition

A Lie group is a manifold G equipped with a group structure where the multiplication and inverse operations are also smooth.

- $\mathbb{R}^n$ and $\mathbb{T}^n$ are Lie groups under standard addition.
- Most matrix groups $GL(V)$, $U(n)$, $SU(n)$, $O(n)$ etc are Lie groups.

Definition

A Lie algebra is a vector space $\mathfrak{g}$ equipped with a bilinear, anti-symmetric binary operation $[\cdot, \cdot]$ called the Lie bracket, that satisfies the Jacobi identity $[x, [y, z]] + [z, [x, y]] + [y, [z, x]] = 0$.

- The vector space $\chi(M)$ of all vector fields on M is a Lie algebra with the Lie bracket $[X, Y]_p f = X_p Y f - Y_p X f$.
- $\chi^{ham}(M)$ and $\chi^{symp}(M)$ form Lie-subalgebras of $\chi(M)$.
- How are Lie algebras and Lie groups related to each other?

Left-invariant vector fields

Definition

*Let G be a Lie group, and L_g be the smooth map that denotes left multiplication by $g \in G$. A vector field X is left-invariant if $(L_g)_*X = X$.*

- The set of left-invariant vector fields is a vector space.

Left-invariant vector fields

Definition

*Let G be a Lie group, and L_g be the smooth map that denotes left multiplication by $g \in G$. A vector field X is left-invariant if $(L_g)_*X = X$.*

- The set of left-invariant vector fields is a vector space.
- If X and Y are left-invariant, then so is $[X, Y] \Rightarrow$ the set of left-invariant vector fields is a Lie-algebra.

Left-invariant vector fields

Definition

*Let G be a Lie group, and L_g be the smooth map that denotes left multiplication by $g \in G$. A vector field X is left-invariant if $(L_g)_*X = X$.*

- The set of left-invariant vector fields is a vector space.
- If X and Y are left-invariant, then so is $[X, Y] \Rightarrow$ the set of left-invariant vector fields is a Lie-algebra.
- There is a natural one-to-one correspondence between left-invariant vector fields, and $T_e G$ with: $X \leftrightarrow X_e$.

Left-invariant vector fields

Definition

*Let G be a Lie group, and L_g be the smooth map that denotes left multiplication by $g \in G$. A vector field X is left-invariant if $(L_g)_*X = X$.*

- The set of left-invariant vector fields is a vector space.
- If X and Y are left-invariant, then so is $[X, Y] \Rightarrow$ the set of left-invariant vector fields is a Lie-algebra.
- There is a natural one-to-one correspondence between left-invariant vector fields, and $T_e G$ with: $X \leftrightarrow X_e$.
- For every Lie group G , $T_e G = \mathfrak{g}$ has a natural Lie algebra structure with the Lie bracket inherited from vector fields!

Left-invariant vector fields

Definition

Let G be a Lie group, and L_g be the smooth map that denotes left multiplication by $g \in G$. A vector field X is left-invariant if $(L_g)_*X = X$.

- The set of left-invariant vector fields is a vector space.
- If X and Y are left-invariant, then so is $[X, Y] \Rightarrow$ the set of left-invariant vector fields is a Lie-algebra.
- There is a natural one-to-one correspondence between left-invariant vector fields, and $T_e G$ with: $X \leftrightarrow X_e$.
- For every Lie group G , $T_e G = \mathfrak{g}$ has a natural Lie algebra structure with the Lie bracket inherited from vector fields!
- Example: The Lie algebra associated with $G = U(n)$ is $\mathfrak{g} = i\mathfrak{H}$, the space of skew-hermitian matrices.

Adjoint and Coadjoint representations

- Let G be a Lie group, and consider the conjugation map $C_g : a \mapsto gag^{-1}$.

Adjoint and Coadjoint representations

- Let G be a Lie group, and consider the conjugation map $C_g : a \mapsto gag^{-1}$.
- The derivative at the identity gives an isomorphism of the Lie algebra $\mathfrak{g}$ with itself.

Adjoint and Coadjoint representations

- Let G be a Lie group, and consider the conjugation map $C_g : a \mapsto gag^{-1}$.
- The derivative at the identity gives an isomorphism of the Lie algebra $\mathfrak{g}$ with itself.
- This gives a natural *representation* of G over its Lie algebra: $Ad : g \mapsto Ad_g = C_{g*} \in GL(\mathfrak{g})$ (*adjoint representation*).

Adjoint and Coadjoint representations

- Let G be a Lie group, and consider the conjugation map $C_g : a \mapsto gag^{-1}$.
- The derivative at the identity gives an isomorphism of the Lie algebra $\mathfrak{g}$ with itself.
- This gives a natural *representation* of G over its Lie algebra:
 $Ad : g \mapsto Ad_g = C_{g*} \in GL(\mathfrak{g})$ (*adjoint representation*).
- It's *dual* representation over $\mathfrak{g}^*$ is given by $Ad_g^*(\zeta)(X) = \zeta(Ad_{g^{-1}}X)$ (*coadjoint representation*).

Adjoint and Coadjoint representations

- Let G be a Lie group, and consider the conjugation map $C_g : a \mapsto gag^{-1}$.
- The derivative at the identity gives an isomorphism of the Lie algebra $\mathfrak{g}$ with itself.
- This gives a natural *representation* of G over its Lie algebra:
 $Ad : g \mapsto Ad_g = C_{g*} \in GL(\mathfrak{g})$ (*adjoint representation*).
- It's *dual* representation over $\mathfrak{g}^*$ is given by $Ad_g^*(\zeta)(X) = \zeta(Ad_{g^{-1}}X)$ (*coadjoint representation*).
- These representations can be interpreted as a G -action on its Lie algebra $\mathfrak{g}$, and allow generalization the theory of Hamiltonian actions.

Moment Maps

Definition

A symplectic action $\psi : G \rightarrow \text{Symp}(M, \omega)$ is Hamiltonian if there exists a map $\mu : M \rightarrow \mathfrak{g}^*$ satisfying:

(i) For each $X \in \mathfrak{g}$, define $\mu^X : p \mapsto \mu_p(X) \in \mathbb{R}$, and let $\tilde{X}$ be the vector field generated on M by the action of the integral curve of X through e in G : $\{\exp(tX) | t \in \mathbb{R}\}$. Then $d\mu^X = i_{\tilde{X}}\omega$.

(ii) $\mu \circ \psi_g = \text{Ad}_g^* \circ \mu$ for all $g \in G$, i.e, μ is equivariant.

- (M, ω, G, μ) is called a *Hamiltonian G -space*, and μ is called the *moment map*.

Moment Maps

Definition

A symplectic action $\psi : G \rightarrow \text{Symp}(M, \omega)$ is Hamiltonian if there exists a map $\mu : M \rightarrow \mathfrak{g}^*$ satisfying:

(i) For each $X \in \mathfrak{g}$, define $\mu^X : p \mapsto \mu_p(X) \in \mathbb{R}$, and let $\tilde{X}$ be the vector field generated on M by the action of the integral curve of X through e in G : $\{\exp(tX) | t \in \mathbb{R}\}$. Then $d\mu^X = i_{\tilde{X}}\omega$.

(ii) $\mu \circ \psi_g = \text{Ad}_g^* \circ \mu$ for all $g \in G$, i.e, μ is equivariant.

- (M, ω, G, μ) is called a *Hamiltonian G -space*, and μ is called the *moment map*.
- This is stronger than the original definition even for $G = \mathbb{R}$ or S^1 (case $X = 1$).

Moment Maps

Definition

A symplectic action $\psi : G \rightarrow \text{Symp}(M, \omega)$ is Hamiltonian if there exists a map $\mu : M \rightarrow \mathfrak{g}^*$ satisfying:

- (i) For each $X \in \mathfrak{g}$, define $\mu^X : p \mapsto \mu_p(X) \in \mathbb{R}$, and let $\tilde{X}$ be the vector field generated on M by the action of the integral curve of X through e in G : $\{\exp(tX) | t \in \mathbb{R}\}$. Then $d\mu^X = i_{\tilde{X}}\omega$.
- (ii) $\mu \circ \psi_g = \text{Ad}_g^* \circ \mu$ for all $g \in G$, i.e. μ is equivariant.

- (M, ω, G, μ) is called a *Hamiltonian G -space*, and μ is called the *moment map*.
- This is stronger than the original definition even for $G = \mathbb{R}$ or S^1 (case $X = 1$).
- Example: The $\mathbb{T}^n$ -action on $\mathbb{C}^n$ given by $(\psi_\alpha(z))_j = e^{ik_j\alpha_j} z_j$, has moment map $(\mu(z))_j = \frac{-1}{2} k_j |z_j|^2 + c_j$.

Moment Maps

Definition

A symplectic action $\psi : G \rightarrow \text{Symp}(M, \omega)$ is Hamiltonian if there exists a map $\mu : M \rightarrow \mathfrak{g}^*$ satisfying:

(i) For each $X \in \mathfrak{g}$, define $\mu^X : p \mapsto \mu_p(X) \in \mathbb{R}$, and let $\tilde{X}$ be the vector field generated on M by the action of the integral curve of X through e in G : $\{\exp(tX) | t \in \mathbb{R}\}$. Then $d\mu^X = i_{\tilde{X}}\omega$.

(ii) $\mu \circ \psi_g = \text{Ad}_g^* \circ \mu$ for all $g \in G$, i.e. μ is equivariant.

- (M, ω, G, μ) is called a *Hamiltonian G -space*, and μ is called the *moment map*.
- This is stronger than the original definition even for $G = \mathbb{R}$ or S^1 (case $X = 1$).
- Example: The $\mathbb{T}^n$ -action on $\mathbb{C}^n$ given by $(\psi_\alpha(z))_j = e^{ik_j\alpha_j}z_j$, has moment map $(\mu(z))_j = \frac{-1}{2}k_j|z_j|^2 + c_j$.
- Example: $U(n)$ acts on $\mathbb{C}^n$ with moment map $\mu(z)(X) = \frac{i}{2}z^*Xz$.

Existence and Uniqueness

- Note that if $H_{\text{de Rham}}^1(M; \mathbb{R}) = 0$, then $\chi^{\text{ham}}(M) = \chi^{\text{symp}}(M)$.

Existence and Uniqueness

- Note that if $H_{\text{de Rham}}^1(M; \mathbb{R}) = 0$, then $\chi^{\text{ham}}(M) = \chi^{\text{symp}}(M)$.
- Motivates the formation of *Lie algebra/Chevalley cohomology groups* $H^k(\mathfrak{g}; \mathbb{R})$ under the linear map $\delta : \Lambda^k \mathfrak{g}^* \rightarrow \Lambda^{k-1} \mathfrak{g}^*$ with $\delta c(X_0, \dots, X_k) = \sum_{i < j} c([X_i, X_j], X_0, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_k)$.

Existence and Uniqueness

- Note that if $H_{\text{de Rham}}^1(M; \mathbb{R}) = 0$, then $\chi^{\text{ham}}(M) = \chi^{\text{symp}}(M)$.
- Motivates the formation of *Lie algebra/Chevalley cohomology groups* $H^k(\mathfrak{g}; \mathbb{R})$ under the linear map $\delta : \Lambda^k \mathfrak{g}^* \rightarrow \Lambda^{k-1} \mathfrak{g}^*$ with $\delta c(X_0, \dots, X_k) = \sum_{i < j} c([X_i, X_j], X_0, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_k)$.
- Every symplectic action is Hamiltonian if $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$.

Existence and Uniqueness

- Note that if $H_{\text{de Rham}}^1(M; \mathbb{R}) = 0$, then $\chi^{\text{ham}}(M) = \chi^{\text{symp}}(M)$.
- Motivates the formation of *Lie algebra/Chevalley cohomology groups* $H^k(\mathfrak{g}; \mathbb{R})$ under the linear map $\delta : \Lambda^k \mathfrak{g}^* \rightarrow \Lambda^{k-1} \mathfrak{g}^*$ with $\delta c(X_0, \dots, X_k) = \sum_{i < j} c([X_i, X_j], X_0, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_k)$.
- Every symplectic action is Hamiltonian if $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$.
- Moment maps are unique for Hamiltonian G -actions if $H^1(\mathfrak{g}; \mathbb{R}) = 0$.

Existence and Uniqueness

- Note that if $H_{\text{de Rham}}^1(M; \mathbb{R}) = 0$, then $\chi^{\text{ham}}(M) = \chi^{\text{symp}}(M)$.
- Motivates the formation of *Lie algebra/Chevalley cohomology groups* $H^k(\mathfrak{g}; \mathbb{R})$ under the linear map $\delta : \Lambda^k \mathfrak{g}^* \rightarrow \Lambda^{k-1} \mathfrak{g}^*$ with $\delta c(X_0, \dots, X_k) = \sum_{i < j} c([X_i, X_j], X_0, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_k)$.
- Every symplectic action is Hamiltonian if $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$.
- Moment maps are unique for Hamiltonian G -actions if $H^1(\mathfrak{g}; \mathbb{R}) = 0$.

Definition

Let G be a compact Lie group with Lie algebra $\mathfrak{g}$. If the commutator ideal $[\mathfrak{g}, \mathfrak{g}] = \{\text{linear combinations of } [X, Y] \text{ for any } X, Y \in \mathfrak{g}\} = \mathfrak{g}$, then G is semisimple.

Existence and Uniqueness

- Note that if $H_{\text{de Rham}}^1(M; \mathbb{R}) = 0$, then $\chi^{\text{ham}}(M) = \chi^{\text{symp}}(M)$.
- Motivates the formation of *Lie algebra/Chevalley cohomology groups* $H^k(\mathfrak{g}; \mathbb{R})$ under the linear map $\delta : \Lambda^k \mathfrak{g}^* \rightarrow \Lambda^{k-1} \mathfrak{g}^*$ with $\delta c(X_0, \dots, X_k) = \sum_{i < j} c([X_i, X_j], X_0, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_k)$.
- Every symplectic action is Hamiltonian if $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$.
- Moment maps are unique for Hamiltonian G -actions if $H^1(\mathfrak{g}; \mathbb{R}) = 0$.

Definition

Let G be a compact Lie group with Lie algebra $\mathfrak{g}$. If the commutator ideal $[\mathfrak{g}, \mathfrak{g}] = \{\text{linear combinations of } [X, Y] \text{ for any } X, Y \in \mathfrak{g}\} = \mathfrak{g}$, then G is semisimple.

- Symplectic actions by compact semisimple Lie groups are Hamiltonian as $H^1(\mathfrak{g}; \mathbb{R}) = [\mathfrak{g}, \mathfrak{g}]^0$.

Existence and Uniqueness

- Note that if $H_{\text{de Rham}}^1(M; \mathbb{R}) = 0$, then $\chi^{\text{ham}}(M) = \chi^{\text{symp}}(M)$.
- Motivates the formation of *Lie algebra/Chevalley cohomology groups* $H^k(\mathfrak{g}; \mathbb{R})$ under the linear map $\delta : \Lambda^k \mathfrak{g}^* \rightarrow \Lambda^{k-1} \mathfrak{g}^*$ with $\delta c(X_0, \dots, X_k) = \sum_{i < j} c([X_i, X_j], X_0, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_k)$.
- Every symplectic action is Hamiltonian if $H^1(\mathfrak{g}; \mathbb{R}) = H^2(\mathfrak{g}; \mathbb{R}) = 0$.
- Moment maps are unique for Hamiltonian G -actions if $H^1(\mathfrak{g}; \mathbb{R}) = 0$.

Definition

Let G be a compact Lie group with Lie algebra $\mathfrak{g}$. If the commutator ideal $[\mathfrak{g}, \mathfrak{g}] = \{\text{linear combinations of } [X, Y] \text{ for any } X, Y \in \mathfrak{g}\} = \mathfrak{g}$, then G is semisimple.

- Symplectic actions by compact semisimple Lie groups are Hamiltonian as $H^1(\mathfrak{g}; \mathbb{R}) = [\mathfrak{g}, \mathfrak{g}]^0$.
- Useful note: For compact and connected Lie groups, $H_{\text{de Rham}}^k(M; \mathbb{R}) = H^k(\mathfrak{g}; \mathbb{R})$.

Table of Contents

- 1 Review of symplectic structure
- 2 Hamiltonian actions
- 3 Generalisation through Lie Theory
- 4 Delzant's Theorem**

Symplectic Reduction

Theorem (Marsden-Weinstein-Meyer)

Let (M, ω, G, μ) be Hamiltonian G -space for a compact Lie group G . Assume G acts freely on $\mu^{-1}(0)$. Then:

- (i) the orbit space $M_{red} = \mu^{-1}(0)/G$ is a manifold.*
- (ii) there is a symplectic form ω_{red} on M_{red} with $i^*\omega = \pi^*\omega_{red}$, where $i : \mu^{-1}(0) \rightarrow M$ is inclusion, and $\pi : \mu^{-1}(0) \rightarrow M_{red}$ is projection.*

Symplectic Reduction

Theorem (Marsden-Weinstein-Meyer)

Let (M, ω, G, μ) be Hamiltonian G -space for a compact Lie group G . Assume G acts freely on $\mu^{-1}(0)$. Then:

- (i) the orbit space $M_{red} = \mu^{-1}(0)/G$ is a manifold.*
- (ii) there is a symplectic form ω_{red} on M_{red} with $i^*\omega = \pi^*\omega_{red}$, where $i : \mu^{-1}(0) \rightarrow M$ is inclusion, and $\pi : \mu^{-1}(0) \rightarrow M_{red}$ is projection.*

- The pair (M_{red}, ω_{red}) is called the *symplectic quotient*.

Symplectic Reduction

Theorem (Marsden-Weinstein-Meyer)

Let (M, ω, G, μ) be Hamiltonian G -space for a compact Lie group G . Assume G acts freely on $\mu^{-1}(0)$. Then:

- (i) the orbit space $M_{red} = \mu^{-1}(0)/G$ is a manifold.*
- (ii) there is a symplectic form ω_{red} on M_{red} with $i^*\omega = \pi^*\omega_{red}$, where $i : \mu^{-1}(0) \rightarrow M$ is inclusion, and $\pi : \mu^{-1}(0) \rightarrow M_{red}$ is projection.*

- The pair (M_{red}, ω_{red}) is called the *symplectic quotient*.
- Example: For the natural action of $U(k)$ on $\mathbb{C}^{k \times n}$, $\mu^{-1}(0) = \{A | AA^\dagger = Id\}$ is freely acted upon, and $\mu^{-1}(0)/G = Gr_{\mathbb{C}}(k, n)$.

Symplectic Reduction

Theorem (Marsden-Weinstein-Meyer)

Let (M, ω, G, μ) be Hamiltonian G -space for a compact Lie group G . Assume G acts freely on $\mu^{-1}(0)$. Then:

- (i) the orbit space $M_{red} = \mu^{-1}(0)/G$ is a manifold.
- (ii) there is a symplectic form ω_{red} on M_{red} with $i^*\omega = \pi^*\omega_{red}$, where $i : \mu^{-1}(0) \rightarrow M$ is inclusion, and $\pi : \mu^{-1}(0) \rightarrow M_{red}$ is projection.

- The pair (M_{red}, ω_{red}) is called the *symplectic quotient*.
- Example: For the natural action of $U(k)$ on $\mathbb{C}^{k \times n}$, $\mu^{-1}(0) = \{A | AA^\dagger = Id\}$ is freely acted upon, and $\mu^{-1}(0)/G = Gr_{\mathbb{C}}(k, n)$.
- Example: For the usual rotation action of S^1 on $\mathbb{C}^{n+1}$, $\mu^{-1}(0) = S^{2n+1}$, and $\mu^{-1}(0)/G \cong \mathbb{CP}^n$, with ω_{red} being exactly the Fubini-Study form.

Symplectic Reduction

- Sketch of proof of (ii):

Symplectic Reduction

- Sketch of proof of (ii):
- Let $\mu_{*,p} : T_p M \rightarrow \mathfrak{g}^*$ be the differential at p of the moment map.

Symplectic Reduction

- Sketch of proof of (ii):
- Let $\mu_{*,p} : T_p M \rightarrow \mathfrak{g}^*$ be the differential at p of the moment map.
- $d\mu = i_{\tilde{X}}\omega \Rightarrow \omega_p(\tilde{X}_p, v) = \mu_{*,p}(v)(X)$ for all $X \in \mathfrak{g}$ and $p \in M$.

Symplectic Reduction

- Sketch of proof of (ii):
- Let $\mu_{*,p} : T_p M \rightarrow \mathfrak{g}^*$ be the differential at p of the moment map.
- $d\mu = i_{\tilde{X}}\omega \Rightarrow \omega_p(\tilde{X}_p, v) = \mu_{*,p}(v)(X)$ for all $X \in \mathfrak{g}$ and $p \in M$.
- Using dimension arguments and statement above, $\ker \mu_{*,p} = (T_p \mathfrak{O}_p)^{\omega_p}$ and $\text{im } \mu_{*,p} = \mathfrak{g}_p^0$, where $\mathfrak{O}_p$ is the G -orbit through p , and $\mathfrak{g}_p$ is the Lie algebra of the stabilizer of p .

Symplectic Reduction

- Sketch of proof of (ii):
- Let $\mu_{*,p} : T_p M \rightarrow \mathfrak{g}^*$ be the differential at p of the moment map.
- $d\mu = i_{\tilde{X}}\omega \Rightarrow \omega_p(\tilde{X}_p, v) = \mu_{*,p}(v)(X)$ for all $X \in \mathfrak{g}$ and $p \in M$.
- Using dimension arguments and statement above, $\ker \mu_{*,p} = (T_p \mathfrak{O}_p)^{\omega_p}$ and $\text{im } \mu_{*,p} = \mathfrak{g}_p^0$, where $\mathfrak{O}_p$ is the G -orbit through p , and $\mathfrak{g}_p$ is the Lie algebra of the stabilizer of p .
- G acts freely on $\mu^{-1}(0) \Rightarrow \mu_*$ is surjective $\Rightarrow \mu^{-1}(0)$ is a closed submanifold $\Rightarrow \ker \mu_{*,p} = T_p \mu^{-1}(0)$.

Symplectic Reduction

- Sketch of proof of (ii):
- Let $\mu_{*,p} : T_p M \rightarrow \mathfrak{g}^*$ be the differential at p of the moment map.
- $d\mu = i_{\tilde{X}}\omega \Rightarrow \omega_p(\tilde{X}_p, v) = \mu_{*,p}(v)(X)$ for all $X \in \mathfrak{g}$ and $p \in M$.
- Using dimension arguments and statement above, $\ker \mu_{*,p} = (T_p \mathfrak{D}_p)^{\omega_p}$ and $\text{im } \mu_{*,p} = \mathfrak{g}_p^0$, where $\mathfrak{D}_p$ is the G -orbit through p , and $\mathfrak{g}_p$ is the Lie algebra of the stabilizer of p .
- G acts freely on $\mu^{-1}(0) \Rightarrow \mu_*$ is surjective $\Rightarrow \mu^{-1}(0)$ is a closed submanifold $\Rightarrow \ker \mu_{*,p} = T_p \mu^{-1}(0)$.
- Equivariance $\Rightarrow \mu^{-1}(0)$ is G -invariant $\Rightarrow T_p \mathfrak{D}_p \subset T_p \mu^{-1}(0) \Rightarrow T_p \mathfrak{D}_p$ is isotropic.

Symplectic Reduction

- Sketch of proof of (ii):
- Let $\mu_{*,p} : T_p M \rightarrow \mathfrak{g}^*$ be the differential at p of the moment map.
- $d\mu = i_{\tilde{X}}\omega \Rightarrow \omega_p(\tilde{X}_p, v) = \mu_{*,p}(v)(X)$ for all $X \in \mathfrak{g}$ and $p \in M$.
- Using dimension arguments and statement above, $\ker \mu_{*,p} = (T_p \mathfrak{D}_p)^{\omega_p}$ and $\text{im } \mu_{*,p} = \mathfrak{g}_p^0$, where $\mathfrak{D}_p$ is the G -orbit through p , and $\mathfrak{g}_p$ is the Lie algebra of the stabilizer of p .
- G acts freely on $\mu^{-1}(0) \Rightarrow \mu_*$ is surjective $\Rightarrow \mu^{-1}(0)$ is a closed submanifold $\Rightarrow \ker \mu_{*,p} = T_p \mu^{-1}(0)$.
- Equivariance $\Rightarrow \mu^{-1}(0)$ is G -invariant $\Rightarrow T_p \mathfrak{D}_p \subset T_p \mu^{-1}(0) \Rightarrow T_p \mathfrak{D}_p$ is isotropic.
- This defines a canonical symplectic form $\omega_{red}([u], [v]) = \omega(u, v)$ on $T_p \mu^{-1}(0) / T_p \mathfrak{D}_p \cong T_{[p]} M_{red}$.

Symplectic Reduction

- Sketch of proof of (ii):
- Let $\mu_{*,p} : T_p M \rightarrow \mathfrak{g}^*$ be the differential at p of the moment map.
- $d\mu = i_{\tilde{X}}\omega \Rightarrow \omega_p(\tilde{X}_p, v) = \mu_{*,p}(v)(X)$ for all $X \in \mathfrak{g}$ and $p \in M$.
- Using dimension arguments and statement above, $\ker \mu_{*,p} = (T_p \mathfrak{D}_p)^{\omega_p}$ and $\text{im } \mu_{*,p} = \mathfrak{g}_p^0$, where $\mathfrak{D}_p$ is the G -orbit through p , and $\mathfrak{g}_p$ is the Lie algebra of the stabilizer of p .
- G acts freely on $\mu^{-1}(0) \Rightarrow \mu_*$ is surjective $\Rightarrow \mu^{-1}(0)$ is a closed submanifold $\Rightarrow \ker \mu_{*,p} = T_p \mu^{-1}(0)$.
- Equivariance $\Rightarrow \mu^{-1}(0)$ is G -invariant $\Rightarrow T_p \mathfrak{D}_p \subset T_p \mu^{-1}(0) \Rightarrow T_p \mathfrak{D}_p$ is isotropic.
- This defines a canonical symplectic form $\omega_{red}([u], [v]) = \omega(u, v)$ on $T_p \mu^{-1}(0) / T_p \mathfrak{D}_p \cong T_{[p]} M_{red}$.
- More generally, shifted moment maps allow for reduction across other levels $\mu^{-1}(\zeta)$ as long as they are preserved under the action.

Symplectic Toric Manifolds

Theorem (Atiyah, Guilleman-Sternberg)

Let (M, ω) be a compact and connected symplectic manifold with a Hamiltonian $\mathbb{T}^m$ action on it with moment map μ . Then the levels of μ are connected, and $\mu(M)$ is the convex hull of the image of fixed points of the action.

Symplectic Toric Manifolds

Theorem (Atiyah, Guilleman-Sternberg)

Let (M, ω) be a compact and connected symplectic manifold with a Hamiltonian $\mathbb{T}^m$ action on it with moment map μ . Then the levels of μ are connected, and $\mu(M)$ is the convex hull of the image of fixed points of the action.

Definition

An effective group action is such that $\bigcap_{p \in M} \text{Stab}(p) = \{e\}$.

Symplectic Toric Manifolds

Theorem (Atiyah, Guilleman-Sternberg)

Let (M, ω) be a compact and connected symplectic manifold with a Hamiltonian $\mathbb{T}^m$ action on it with moment map μ . Then the levels of μ are connected, and $\mu(M)$ is the convex hull of the image of fixed points of the action.

Definition

An effective group action is such that $\bigcap_{p \in M} \text{Stab}(p) = \{e\}$.

- A smooth effective $\mathbb{T}^m$ action is a submersion at some point p , and thus must have at least $m + 1$ fixed points.

Symplectic Toric Manifolds

Theorem (Atiyah, Guilleman-Sternberg)

Let (M, ω) be a compact and connected symplectic manifold with a Hamiltonian $\mathbb{T}^m$ action on it with moment map μ . Then the levels of μ are connected, and $\mu(M)$ is the convex hull of the image of fixed points of the action.

Definition

An effective group action is such that $\cap_{p \in M} \text{Stab}(p) = \{e\}$.

- A smooth effective $\mathbb{T}^m$ action is a submersion at some point p , and thus must have at least $m + 1$ fixed points.
- As $T_p \mathfrak{D}_p$ is isotropic, $\dim M \geq 2m$.

Symplectic Toric Manifolds

Definition

A symplectic toric manifold is a compact and connected symplectic manifold (M, ω) , equipped with a $\mathbb{T}^m$ action with $m = \frac{1}{2} \dim M$.

Symplectic Toric Manifolds

Definition

A symplectic toric manifold is a compact and connected symplectic manifold (M, ω) , equipped with a $\mathbb{T}^m$ action with $m = \frac{1}{2}\dim M$.

- Example: S^1 acting on S^2 by rotations with moment map $\mu = h$ is a symplectic toric manifold with $\mu(S^2) = [-1, 1]$.

Symplectic Toric Manifolds

Definition

A symplectic toric manifold is a compact and connected symplectic manifold (M, ω) , equipped with a $\mathbb{T}^m$ action with $m = \frac{1}{2} \dim M$.

- Example: S^1 acting on S^2 by rotations with moment map $\mu = h$ is a symplectic toric manifold with $\mu(S^2) = [-1, 1]$.
- $\mathbb{CP}^2$ is toric under the Fubini-Study structure and action $(e^{i\theta_1}, e^{i\theta_2})[z_0, z_1, z_2] = [z_0, e^{i\theta_1} z_1, e^{i\theta_2} z_2]$ with moment map $\mu([z]) = \frac{-1}{2} \left(\frac{|z_1|^2}{|z|^2}, \frac{|z_2|^2}{|z|^2} \right)$. $\mu(\mathbb{CP}^2) = ?$

Symplectic Toric Manifolds

Definition

A symplectic toric manifold is a compact and connected symplectic manifold (M, ω) , equipped with a $\mathbb{T}^m$ action with $m = \frac{1}{2} \dim M$.

- Example: S^1 acting on S^2 by rotations with moment map $\mu = h$ is a symplectic toric manifold with $\mu(S^2) = [-1, 1]$.
- $\mathbb{CP}^2$ is toric under the Fubini-Study structure and action $(e^{i\theta_1}, e^{i\theta_2})[z_0, z_1, z_2] = [z_0, e^{i\theta_1} z_1, e^{i\theta_2} z_2]$ with moment map $\mu([z]) = \frac{-1}{2} \left(\frac{|z_1|^2}{|z|^2}, \frac{|z_2|^2}{|z|^2} \right)$. $\mu(\mathbb{CP}^2) = ?$
- What about $\mathbb{CP}^3$ under a similar action?

Symplectic Toric Manifolds

Definition

A symplectic toric manifold is a compact and connected symplectic manifold (M, ω) , equipped with a $\mathbb{T}^m$ action with $m = \frac{1}{2} \dim M$.

- Example: S^1 acting on S^2 by rotations with moment map $\mu = h$ is a symplectic toric manifold with $\mu(S^2) = [-1, 1]$.
- $\mathbb{CP}^2$ is toric under the Fubini-Study structure and action $(e^{i\theta_1}, e^{i\theta_2})[z_0, z_1, z_2] = [z_0, e^{i\theta_1} z_1, e^{i\theta_2} z_2]$ with moment map $\mu([z]) = \frac{-1}{2} \left(\frac{|z_1|^2}{|z|^2}, \frac{|z_2|^2}{|z|^2} \right)$. $\mu(\mathbb{CP}^2) = ?$
- What about $\mathbb{CP}^3$ under a similar action?
- An observation: Dimension of the face + Dimension of the stabilizer = m .

Delzant Polytopes

Definition

A Delzant polytope $\triangle$ in $\mathbb{R}^n$ is a convex polytope that is:

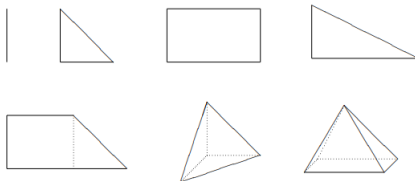
- (i) simple, i.e, has n edges meeting each vertex;
- (ii) rational, i.e, each edge meeting at a vertex p is of the form $p + tu_i$ where $u_i \in \mathbb{Z}^n$;
- (iii) smooth, i.e, for each vertex p , $\{u_i\}$ forms a $\mathbb{Z}$ -basis of $\mathbb{Z}^n$.

Delzant Polytopes

Definition

A Delzant polytope Δ in $\mathbb{R}^n$ is a convex polytope that is:

- (i) simple, i.e. has n edges meeting each vertex;
- (ii) rational, i.e. each edge meeting at a vertex p is of the form $p + tu_i$ where $u_i \in \mathbb{Z}^n$;
- (iii) smooth, i.e. for each vertex p , $\{u_i\}$ forms a $\mathbb{Z}$ -basis of $\mathbb{Z}^n$.



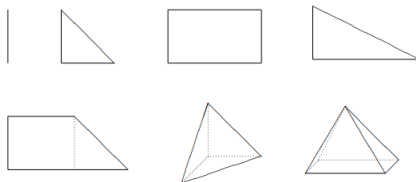
- Which of the above are Delzant polytopes?

Delzant Polytopes

Definition

A Delzant polytope Δ in $\mathbb{R}^n$ is a convex polytope that is:

- (i) simple, i.e, has n edges meeting each vertex;
- (ii) rational, i.e, each edge meeting at a vertex p is of the form $p + tu_i$ where $u_i \in \mathbb{Z}^n$;
- (iii) smooth, i.e, for each vertex p , $\{u_i\}$ forms a $\mathbb{Z}$ -basis of $\mathbb{Z}^n$.



- Which of the above are Delzant polytopes?
- Any Delzant polytope can be written as the intersection of half-spaces $\Delta = \{x \in \mathbb{R}^n \mid \langle x, v_i \rangle \leq \lambda_i\}$ where $\lambda_i \in \mathbb{R}$ and v_i are outward pointing vectors (usually chosen to be primitive) to each *facet*.

Delzant's Theorem

Theorem (Delzant, 1988)

There is a one-to-one correspondence between symplectic toric manifolds and Delzant polytopes:

$$(M^{2n}, \omega, \mathbb{T}^n, \mu) \leftrightarrow \mu(M).$$

- We shall sketch the proof of the surjectivity.

Delzant's Theorem

Theorem (Delzant, 1988)

There is a one-to-one correspondence between symplectic toric manifolds and Delzant polytopes:

$$(M^{2n}, \omega, \mathbb{T}^n, \mu) \leftrightarrow \mu(M).$$

- We shall sketch the proof of the surjectivity.
- Motivation: Every n -dimensional Delzant polytope with d facets can be formed by the intersection of an affine plane with the negative orthant $\mathbb{R}_-^d$.
- Idea: For an n -dimensional Delzant polytope with d facets, we show that there is symplectic quotient of $(\mathbb{C}^d, \omega_0, \mathbb{T}^d, \mu)$ with action $(e^{2\pi i t_1}, \dots, e^{2\pi i t_d}) \cdot (z_1, \dots, z_d) = (e^{2\pi i t_1} z_1, \dots, e^{2\pi i t_d} z_d)$ and moment map $\phi(z_1, \dots, z_d) = -\pi(|z_1|^2, \dots, |z_d|^2) + (\lambda_1, \dots, \lambda_d)$ that does the job.

The Delzant Construction

- Let $\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\}$ and $\{e_i\}$ be standard basis of $\mathbb{R}^d$.

The Delzant Construction

- Let $\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\}$ and $\{e_i\}$ be standard basis of $\mathbb{R}^d$.
- Define the surjective map $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ via $\pi(e_i) = v_i$, with $\pi(\mathbb{Z}^d) = \mathbb{Z}^n$.

The Delzant Construction

- Let $\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\}$ and $\{e_i\}$ be standard basis of $\mathbb{R}^d$.
- Define the surjective map $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ via $\pi(e_i) = v_i$, with $\pi(\mathbb{Z}^d) = \mathbb{Z}^n$.
- Pull π down to a map $\mathbb{T}^d \rightarrow \mathbb{T}^n$. Let $N = \ker \pi$ be a $(d - n)$ dimensional sub-torus.

The Delzant Construction

- Let $\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\}$ and $\{e_i\}$ be standard basis of $\mathbb{R}^d$.
- Define the surjective map $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ via $\pi(e_i) = v_i$, with $\pi(\mathbb{Z}^d) = \mathbb{Z}^n$.
- Pull π down to a map $\mathbb{T}^d \rightarrow \mathbb{T}^n$. Let $N = \ker \pi$ be a $(d - n)$ dimensional sub-torus.
- Induces an exact sequence $0 \rightarrow N \xrightarrow{i} \mathbb{T}^d \xrightarrow{\pi} \mathbb{T}^n \rightarrow 0$.

The Delzant Construction

- Let $\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\}$ and $\{e_i\}$ be standard basis of $\mathbb{R}^d$.
- Define the surjective map $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ via $\pi(e_i) = v_i$, with $\pi(\mathbb{Z}^d) = \mathbb{Z}^n$.
- Pull π down to a map $\mathbb{T}^d \rightarrow \mathbb{T}^n$. Let $N = \ker \pi$ be a $(d - n)$ dimensional sub-torus.
- Induces an exact sequence $0 \rightarrow N \xrightarrow{i} \mathbb{T}^d \xrightarrow{\pi} \mathbb{T}^n \rightarrow 0$.
- Induces a dual exact sequence $0 \rightarrow (\mathbb{R}^n)^* \xrightarrow{\pi^*} (\mathbb{R}^d)^* \xrightarrow{i^*} \mathfrak{n}^* \rightarrow 0$.

The Delzant Construction

- Let $\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\}$ and $\{e_i\}$ be standard basis of $\mathbb{R}^d$.
- Define the surjective map $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ via $\pi(e_i) = v_i$, with $\pi(\mathbb{Z}^d) = \mathbb{Z}^n$.
- Pull π down to a map $\mathbb{T}^d \rightarrow \mathbb{T}^n$. Let $N = \ker \pi$ be a $(d - n)$ dimensional sub-torus.
- Induces an exact sequence $0 \rightarrow N \xrightarrow{i} \mathbb{T}^d \xrightarrow{\pi} \mathbb{T}^n \rightarrow 0$.
- Induces a dual exact sequence $0 \rightarrow (\mathbb{R}^n)^* \xrightarrow{\pi^*} (\mathbb{R}^d)^* \xrightarrow{i^*} \mathfrak{n}^* \rightarrow 0$.
- Sub-action of N is Hamiltonian with moment map $i^* \circ \phi$.

The Delzant Construction

- Let $\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\}$ and $\{e_i\}$ be standard basis of $\mathbb{R}^d$.
- Define the surjective map $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ via $\pi(e_i) = v_i$, with $\pi(\mathbb{Z}^d) = \mathbb{Z}^n$.
- Pull π down to a map $\mathbb{T}^d \rightarrow \mathbb{T}^n$. Let $N = \ker \pi$ be a $(d - n)$ dimensional sub-torus.
- Induces an exact sequence $0 \rightarrow N \xrightarrow{i} \mathbb{T}^d \xrightarrow{\pi} \mathbb{T}^n \rightarrow 0$.
- Induces a dual exact sequence $0 \rightarrow (\mathbb{R}^n)^* \xrightarrow{\pi^*} (\mathbb{R}^d)^* \xrightarrow{i^*} \mathfrak{n}^* \rightarrow 0$.
- Sub-action of N is Hamiltonian with moment map $i^* \circ \phi$.
- $Z = (i^* \circ \phi)^{-1}(0)$ is a $(d + n)$ -dimensional compact submanifold of $\mathbb{C}^d$ and N acts freely on Z .

The Delzant Construction

- Let $\Delta = \{x \in (\mathbb{R}^n)^* \mid \langle x, v_i \rangle \leq \lambda_i, \ i = 1, \dots, d\}$ and $\{e_i\}$ be standard basis of $\mathbb{R}^d$.
- Define the surjective map $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^n$ via $\pi(e_i) = v_i$, with $\pi(\mathbb{Z}^d) = \mathbb{Z}^n$.
- Pull π down to a map $\mathbb{T}^d \rightarrow \mathbb{T}^n$. Let $N = \ker \pi$ be a $(d - n)$ dimensional sub-torus.
- Induces an exact sequence $0 \rightarrow N \xrightarrow{i} \mathbb{T}^d \xrightarrow{\pi} \mathbb{T}^n \rightarrow 0$.
- Induces a dual exact sequence $0 \rightarrow (\mathbb{R}^n)^* \xrightarrow{\pi^*} (\mathbb{R}^d)^* \xrightarrow{i^*} \mathfrak{n}^* \rightarrow 0$.
- Sub-action of N is Hamiltonian with moment map $i^* \circ \phi$.
- $Z = (i^* \circ \phi)^{-1}(0)$ is a $(d + n)$ -dimensional compact submanifold of $\mathbb{C}^d$ and N acts freely on Z .
- Using Marsden-Weinstein-Meyer reduction, we obtain a symplectic quotient $(M_\Delta, \omega_\Delta)$, which is a Hamiltonian $\mathbb{T}^n$ space.

Example

- Consider the Delzant polytope $\triangle = [0, a] \subset \mathbb{R}^*$ with $n = 1$ and $d = 2$, $v_1 = 1$ and $v_2 = -1$.

Example

- Consider the Delzant polytope $\triangle = [0, a] \subset \mathbb{R}^*$ with $n = 1$ and $d = 2$, $v_1 = 1$ and $v_2 = -1$.
- $\triangle$ is defined by $\langle x, v_1 \rangle \leq 0$ and $\langle x, v_2 \rangle \leq a$.

Example

- Consider the Delzant polytope $\Delta = [0, a] \subset \mathbb{R}^*$ with $n = 1$ and $d = 2$, $v_1 = 1$ and $v_2 = -1$.
- Δ is defined by $\langle x, v_1 \rangle \leq 0$ and $\langle x, v_2 \rangle \leq a$.
- $\pi(e_1) = 1$ and $\pi(e_2) = -1 \Rightarrow \ker \pi = \text{span}(e_1 + e_2) \Rightarrow N \cong S^1$.

Example

- Consider the Delzant polytope $\triangle = [0, a] \subset \mathbb{R}^*$ with $n = 1$ and $d = 2$, $v_1 = 1$ and $v_2 = -1$.
- $\triangle$ is defined by $\langle x, v_1 \rangle \leq 0$ and $\langle x, v_2 \rangle \leq a$.
- $\pi(e_1) = 1$ and $\pi(e_2) = -1 \Rightarrow \ker \pi = \text{span}(e_1 + e_2) \Rightarrow N \cong S^1$.
- $i^*(x_1, x_2) = x_1 + x_2$ and $i^* \circ \phi(z_1, z_2) = -\pi(|z_1|^2 + |z_2|^2) + a$.

Example

- Consider the Delzant polytope $\triangle = [0, a] \subset \mathbb{R}^*$ with $n = 1$ and $d = 2$, $v_1 = 1$ and $v_2 = -1$.
- $\triangle$ is defined by $\langle x, v_1 \rangle \leq 0$ and $\langle x, v_2 \rangle \leq a$.
- $\pi(e_1) = 1$ and $\pi(e_2) = -1 \Rightarrow \ker \pi = \text{span}(e_1 + e_2) \Rightarrow N \cong S^1$.
- $i^*(x_1, x_2) = x_1 + x_2$ and $i^* \circ \phi(z_1, z_2) = -\pi(|z_1|^2 + |z_2|^2) + a$.
- $(i^* \circ \phi)^{-1}(0) = S_{\frac{a}{\pi}}^2$.

Example

- Consider the Delzant polytope $\Delta = [0, a] \subset \mathbb{R}^*$ with $n = 1$ and $d = 2$, $v_1 = 1$ and $v_2 = -1$.
- Δ is defined by $\langle x, v_1 \rangle \leq 0$ and $\langle x, v_2 \rangle \leq a$.
- $\pi(e_1) = 1$ and $\pi(e_2) = -1 \Rightarrow \ker \pi = \text{span}(e_1 + e_2) \Rightarrow N \cong S^1$.
- $i^*(x_1, x_2) = x_1 + x_2$ and $i^* \circ \phi(z_1, z_2) = -\pi(|z_1|^2 + |z_2|^2) + a$.
- $(i^* \circ \phi)^{-1}(0) = S_{\frac{a}{\pi}}^2$.
- $(M_\Delta, \omega_\Delta) = (\mathbb{CP}^1, \omega_{FS})$, has a standard toric structure under rotational action of $\mathbb{T}^2/N = S^1$.

Example

- Consider the Delzant polytope $\Delta = [0, a] \subset \mathbb{R}^*$ with $n = 1$ and $d = 2$, $v_1 = 1$ and $v_2 = -1$.
- Δ is defined by $\langle x, v_1 \rangle \leq 0$ and $\langle x, v_2 \rangle \leq a$.
- $\pi(e_1) = 1$ and $\pi(e_2) = -1 \Rightarrow \ker \pi = \text{span}(e_1 + e_2) \Rightarrow N \cong S^1$.
- $i^*(x_1, x_2) = x_1 + x_2$ and $i^* \circ \phi(z_1, z_2) = -\pi(|z_1|^2 + |z_2|^2) + a$.
- $(i^* \circ \phi)^{-1}(0) = S_{\frac{a}{\pi}}^2$.
- $(M_\Delta, \omega_\Delta) = (\mathbb{CP}^1, \omega_{FS})$, has a standard toric structure under rotational action of $\mathbb{T}^2/N = S^1$.
- Neat observation: Consider a face F of $\phi(Z)$ of dimension $n - r$ dimension of the stabilizer of a point $z \in Z$ with image in F has dimension r .

Thank you![2][1][3]



V.I Arnold.

Mathematical Methods of Classical Mechanics.

Springer-Verlag, 1974.



Annas Cannas de Silva.

Lectures on Symplectic Geometry, volume 1764 of *Lecture Notes in Mathematics*.

Springer-Verlag, 2001.



Jr. Kirillov, Alexander.

Introduction to Lie Groups and Lie Algebras.

Cambridge Studies in Advanced Mathematics, 2008.